Attention Legos

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1 author:

Gary Nan Tie
Mu Risk LLC

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Abstract
Attention building blocks for machines to learn world models?

Introduction
Humans have five basic senses: sight, hearing, touch, taste, and smell. These senses help us perceive and interact with the environment around us, and in doing so, we each create a world model, our representation of reality.

Analogously, in machine learning, large language models are trained on immense corpora of data through transformer attention mechanisms. Soft attention is a mechanism in machine learning that allows models to focus on different parts of the input data by assigning varying levels of importance to each element. Finding efficient ways to search the corresponding token space for continuation or translation is paramount. There are many attention mechanisms based on the query-key-value database retrieval paradigm. However, all these statistical approaches do not take into account language meaning embodied by key-value semantics. More generally, multimedia attention mechanisms are needed.

Liftings of certain fibrations on the other hand, generate query updates that respect key-value semantics. Moreover, fibration structure [1] enables natural expression of fundamental attention mechanisms that can adapt to the diverse features of multimedia. For example, query context can be taken into account, or the set of keys can be enlarged with similar ones to extend key-value semantics. In addition, we can mix and match such building blocks, as well as concatenating them in parallel, to architect attention as desired.

We introduce ‘attention lego’ [13], primary structures to systematically design bespoke multimedia attention architectures.

Query update legos:

S. Similarity function

F. Fibration lifting
- **M.** Multi-head
- **Q.** Query context
- **E.** Extended key-value semantics

Set up
We are given queries q_i ε Q, i ε [n] and a key-value pairing p: K -> V, such that $p(k_j) = v_j$, j ε [m]. We want to update and refine queries until perplexity is acceptable.

S. Similarity function
Generic transform attention mechanisms (for sequence-to-sequence or next-token) apply a feed forward neural network element wise to query updates to introduce non-linearity in refinement. However, such statistical approaches ignore language semantics.

Let score be similarity function, like a reproducing kernel. Let probability α_{ij} = softmax score(q_i, k_j) and let j^* = arg max α_{ij}.
Let initial query update
$q_i' = \Sigma_j \alpha_{ij} v_j$, then define query update = FFN[q_i'],
where FFN is a feed forward network applied element wise.
The unconstrained learning of the FFN can lead to hallucinations.

F. Fibration lifting [12, 10, 9]
In contrast, we can learn a fibration lifting neural network that simultaneously updates a query and abductively finds a key to explain why the output value was returned. This mathematical coherence respects and takes into account the semantics of the key-value pairing.

Let m: Q -> Q' denote query updates and p: K -> V
a key-value pairing. We construct a square diagram from:
m x id_K: Q x K -> Q' x K and $id_{Q'}$ x p : Q' x K -> Q' x V,
id_Q x p : Q x K -> Q x V and m x id_V : Q x V -> Q' x V.
Suppose the diagram commutes, that is
($id_{Q'}$ x p) o (m x id_K) = (m x id_V) o (id_Q x p).

With $m(qi) = qi'$ or a prior update, learn diagonal lifting
$\sigma : Q \times V \to Q' \times K$ so that both triangles commute,
that is $\sigma \circ (idQ \times p) = m \times idK$ and $(idQ' \times p) \circ \sigma = m \times idV$.
Which is to say $idQ' \times p$ is fibration, see the references for details. Define fibration query update to be:
$qi^* = proj1\ \sigma(m(qi), vj^*)$, where $vj^* = p(kj^*)$.
So instead of applying a feed forward network, we learn a fibration lifting network that takes into account language semantics to update queries.

Fibration diagrams can be customized for different purposes such as:

M. Multi-head [14]
Multimedia combines different content forms such as text, audio, images, animations, video, or haptics, necessitating different head semantics. In each head, an input query is initially updated by **S,** then in parallel multiple **F** query updates are performed, each with their own key-value pairing semantics, as well as with concatenations for refinement if necessary. These multiple parallel updates can be gated into a final updated output query, from which the key-value pairings determine the best continuation output value.

Q. Query context [15]
A window of shoulder queries around an input query contains contextual information. Each query in the window can be updated by **F** and **S** and concatenations thereof, weighted by a prior distribution, so that similarity, semantics and context are all taken into account.

E. Extended key-value semantics [16]
Query updates of a fibration lifting are determined by semantics $p: K \to V$. We can enlarge K by carefully adding highly similar keys, so abduction has more keys to choose from to explain a value, thereby enhancing semantics and abductive inference.

Conclusion
The five building blocks **S, F, M, Q**, **E** enable purpose specific transformer attention mechanisms to be designed on multimedia input with tailor made semantics. For example, CGAN + Fibration Attention **F,** to generate more plausible images. This is a blueprint for attention legos, and even after designing a domain specific attention architecture, there is much engineering and training to be done for a functioning LLM!

Machine translation via transformer attention **S,** had been trained to translate multiple language pairs, like English to Japanese and English to Korean, and then something unexpected happened. Researchers realized it could also translate between Japanese and Korean, even though it had never been directly taught how to. The reason? The neural network had invented its own interlingua, a kind of internal language that helped it understand meanings instead of just memorizing words. It is extraordinary that soft attention coaxed out the illusion of comprehension from swaths of syntactical examples, contextual associations conveyed meaning, however pattern motifs can become jumbled like when LLMs hallucinate.

In the same vein, as the fibration structure of **F** introduces semantics and abductive inference to attention, it provides a new dimension for emergent behavior. The fibration attention mechanisms **F, M, Q, E** for the first time enable LLM to think abductively. This is like a doctor making a medical diagnosis. We know diseases can cause certain symptoms. In reverse, observing patient symptoms and hypothesizing which diseases could explain them, is abductive inference. So beyond statistical pattern recognition, fibration attention enables idea discovery. Maybe machines will invent their own internal world models using abductive attention legos? Recursive loops of attention blocks have the potential for self learning, perhaps generating a machine world model through its explorations of multimedia data via these five primary attention senses. The ability to manipulate meaning, hypothesize explanations and discover new ideas surely puts us at the threshold of artificial general intelligence. However, neither human nor machine world views can be appreciated by the other, unless somehow one gets to experience what the other does! A brave new world.

References (most recent first)

[16] ‘Deep Fibration Attention’
Gary Nan Tie, Aug 31, 2025
DOI: 10.13140/RG.2.2.17946.30402
In fibration attention, a lifting simultaneously updates a query and abductively finds a key to explain a value. This all depends on the key-value pairing (k,v). By determining other keys that are highly similar to k, to also explain v, we can enhance key-value semantics to achieve more meaningful attention. Moreover, we can recursively layer keys upon keys in a tree, to deepen the set of value explanations.

[15] ‘Contextual Fibration Attention’
Gary Nan Tie, Aug 29,2025
DOI: 10.13140/RG.2.2.12205.35044
In a query-key-value attention model, the key-value pairing conveys language semantics. Query context also carries meaning. We introduce an abductive fibration attention model that holistically captures both. A fibration lifting, jointly learns from query context and language semantics (that abductively finds a key to best explain the returned value).

[14] ‘Multi-head abductive attention’
Gary Nan Tie, Aug 25, 2025
DOI: 10.13140/RG.2.2.20597.23521
Transformer multi-head attention can focus on different parts of an input query sequence, however without regard to potentially different key-value semantics. This new primitive for formal query updating, constructs updates to a query in parallel, by learning multi-head abductive fibration liftings, to refine similarity based attention, utilizing the semantics of multiple key-value pairings.

[13] ‘Formal query updating to generate problem specific fibration abduction attention algorithms’
Gary Nan Tie, Aug 23, 2025
DOI: 10.13140/RG.2.2.33029.41446
A formalism to construct problem specific query update algorithms founded on standard attention and fibration attention primitives.

[12] ‘Fibformer - a fibration transformer’
Gary Nan Tie, Aug 22, 2025
DOI: 10.13140/RG.2.2.36293.10723
We introduce a new transformer with novel fibration attention based on abductive reasoning. This coherent approach to query updating enables controlled structured exploration of continuation values.

[11] ‘LLM Bonsai - pruning FibChat
Gary Nan Tie, Jul 25, 2025
DOI: 10.13140/RG.2.2.21984.80644
Each step in learning the diagonal lifting of a fibration, used to simultaneously update a query and find a section of the key-value pairing, represents an opportunity for intervention to direct the growth of our query refinement tree by selective pruning, so as to avoid hallucinations, LLM bonsai!

[10] ‘FibChat - It makes stuff up, but with structure.’
Gary Nan Tie · July 2025
DOI: 10.13140/RG.2.2.10549.59364/2
FibChat is a LLM based on fibration attention, that enables coherent structured mitigation of hallucinations.

[9] ‘Attention Fibration Abduction’
Gary Nan Tie, Jul 2, 2025
DOI: 10.13140/RG.2.2.29060.23680/1
Transformer attention and abductive reasoning are intimately connected, embodied by fibrations. Suitable attention mechanisms can make a key-value pairing a learnable neural network fibration for abductive reasoning. A fibration lifting explains an attention output value to a query, by identifying the key paired to it, explainable attention through abduction.

[8] ‘Parsimonious Neural Network Abduction’
Gary Nan Tie, Jun 22, 2025
DOI: 10.13140/RG.2.2.15695.80804
For hypotheses whose effect is manifested by observations, abduction seeks to explain a given observation by finding a hypothesis whose effect is that observation. Abductive reasoning has a fibration semantics that can be implemented by neural networks; a step towards artificial general intelligence. In this note we introduce a parsimonious choice of explanation depending on one’s utility function.

[7] ‘Neural Nerwork Abduction’
Gary Nan Tie, Jun 13, 2025
DOI: 10.13140/RG.2.2.18506.07360/1
For hypotheses whose effect is manifested by observations, abduction seeks to explain a given observation by finding a hypothesis whose effect is that observation. Abductive reasoning has a fibration semantics that can be implemented by neural networks; a step towards artificial general intelligence.
A natural application is in making a medical diagnosis. Fibration structure and properties enable coherency and consistency in this process.
Moreover being machine learnable, can help physicians uncover diagnostic insights in large datasets.

[6] ‘Meta-Abduction’
Gary Nan Tie, Jun 10, 2025
DOI: 10.13140/RG.2.2.22022.64323
We note that for fibration semantics of abductive reasoning, abduction of attention is a meta-effect, not section reversal per se.

[5] ‘Abductive Inference via Explainable Neural Networks’
Gary Nan Tie, May 25, 2025
DOI: 10.13140/RG.2.2.20953.84320
Explainable abductive inference by invertible neural networks is examined in the context of fibration semantics.

[4] ‘Abductive Reasoning Precedence’
Gary Nan Tie, May 23, 2025
DOI: 10.13140/RG.2.2.19913.45923
Fibration liftings used to model abduction are not necessarily unique. So chains of abductive reasoning are a series of certain choices of liftings. To keep track of this precedence we introduce a simple mechanism to mark liftings. This transparency enables one to analyze decision points in an exploit-explore strategy for optimizing chains of abductive reasoning.

[3] ‘Coherent Abductive Medical Diagnosis’
Gary Nan Tie, May 15, 2025
DOI: 10.13140/RG.2.2.16148.41605
Disease causes symptoms. In the other direction, given a patient’s symptoms, medical diagnosis seeks diseases that explain the observed symptoms, a form of abductive reasoning. Abduction can be viewed as a fibration, a mathematical model, which when applied to clinical decision making enables coherent, transparent, refinable medical diagnoses. Moreover, given enough data, fibrations are potentially machine learnable, providing an interactive AI learning tool for physicians to better understand and refine their diagnostic thinking.

[2] ‘Fibration Abductive Learning’
Gary Nan Tie, Mar 4, 2025
DOI: 10.13140/RG.2.2.27284.82562/2
Neural networks like MLP (multilayer perceptrons) and KAN (Kolmogorov-Arnold networks) can solve, by induction, regression, classification and time series problems. Deduction is used by neural networks in rule based systems, logic programming, automated theorem proving and reinforcement learning. This naturally begs the question about applying the third kind of inference abduction, seeking the simplest and most likely explanation for a set of observations. To conceptualize and perform abductive machine learning, categorical fibrations are proposed, as they coherently give both context and structure for abductive reasoning via a tautological attention mechanism.

[1] ‘Fibrations explain all you need!
- Fibrations, Abduction, Attention
Gary Nan Tie, Mar 4, 2025
DOI: 10.13140/RG.2.2.35984.52488
To conceptualize and perform abductive machine learning, categorical fibrations are proposed, as they coherently give both context and structure for abductive reasoning via an attention mechanism. Moreover, 2-categorical lifting compatibility conditions ensure consistent hierarchical and parallel explanations.